



Management Performance:

Pol 06 Reduction of nighttime light pollution



Credits

No Minimum Standard

Aim

To ensure that external lighting is concentrated in the appropriate areas and that upward lighting is minimised, thereby reducing unnecessary light pollution and nuisance to neighbours.

Question

Has an assessment been performed that demonstrates that external lighting does not result in light pollution at night?

Credits	Answer	Select a single answer option
0	A.	Question not answered
0	B.	No
2	C.	Yes, qualitative assessment performed
4	D.	Yes, quantitative assessment performed by a lighting professional
4	E.	Exterior lighting is not provided, and it is not required from a safety perspective

Assessment criteria

Criterion	Assessment criteria	Applicable Answer
1.	For qualitative assessments where external lighting is present: a) All external lighting is aimed downwards, to avoid upward sky glow. b) All external lighting is aimed away from neighbouring buildings or open spaces or has screens or baffles to prevent light spill to them. c) There are no illuminated signs. d) Lighting is automatically controlled to be switched off or dimmed to a much lower level between 23:00 and 07:00 (the curfew period).	C
2.	For quantitative assessments where external lighting is present, a lighting professional confirms: a) All external lighting (except for safety and security lighting) complies with the pre-curfew limitations in Table 36 in the Checklists and Tables section and can be automatically switched off between 23:00 and 07:00 (the curfew period).	D

Criterion	Assessment criteria	Applicable Answer
	b) Illuminated signs, where present, comply with the maximum luminance (cd/m²) in Table 36. c) If safety or security lighting is installed and is used between 23:00 and 07:00 it complies with the post-curfew limitations in Table 36 (e.g. by using an automatic switch to reduce the lighting levels at 23:00 or earlier). d) Where non-security lighting is essential between 23:00 and 07:00, i.e. for buildings which open or operate between these times, the lighting system can automatically switch to the lighting levels for post-curfew in Table 36.	
3.	A lighting professional confirms that external obtrusive lighting has been eliminated through effective design that removes the need for external lighting without adversely affecting the safety and security of the site and its users.	E
4.	Assessment of illuminance or luminous intensity is not required if all luminaires are cut-off types and angled so that light in potentially obtrusive directions is blocked. Assessment of upward light ratio is not required if all luminaires are cut-off types delivering downward light only.	C - E
5.	Flush stud lights used for safety purposes in vehicle manoeuvring areas may be excluded from the assessment.	C - E
6.	All types of illuminated sign must meet the criteria, both self-illuminated and those illuminated by reflection from other sources.	C - E
7.	For the purposes of this issue, external lighting includes both lighting mounted externally, and lighting mounted inside an asset that is primarily intended to enhance its external appearance or light external spaces after dark.	C - E

Specific notes

Asset type specific	
1.	Different Curfew times Where a different curfew time applies for other reasons (e.g. noise control), consideration should be given to the coordination of the curfews, e.g. allowing sufficient time of operation for the lighting after the conclusion of the activity to facilitate crowd dispersal, particularly where large numbers of spectators are involved.
2.	Security lighting Where light fittings are specified to comply with specific security standards and these conflict with the criteria, they can be excluded from assessment. In these circumstances, the Assessor

Asset type specific	
	must obtain evidence confirming the specific security standards and that they are applicable to the asset.

Methodology

Quantitative assessment by lighting professional

The direct measurement of upward light ratio (ULR), vertical illuminance and luminous intensity may not be possible due to issues with access to luminaires or to neighbouring properties. Consequently, the lighting professional may use their professional judgement to determine compliance with the requirements in Table 36, e.g. visual assessment to estimate the proportion of ULR, or where access to sensitive receptors is not possible, luminous intensity measurements could be made from accessible locations in the direction of the view of luminaires as seen from sensitive receptors. Where lighting design documentation is available from when the lighting was installed, this may be used to demonstrate compliance with the assessment issue criteria.

Checklists and Tables

Table 36: Obtrusive light limitations for exterior lighting installations and illuminated signs

Zone (see Table 37)	Maximum luminaire upward light ratio (ULR) / %	Maximum vertical illuminance on nearby properties' windows (E _v) / lux		Maximum luminous intensity emitted by luminaire (I) / cd		Maximum average surface luminance of building façade (L)* / cd/m ²	Maximum average surface luminance of signs (L)* / cd/m ²
		Pre-curfew	Post-curfew	Pre-curfew	Post-curfew		
E0	0	0	0	0	0	<0.1	<0.1
E1	0	2	0 [#]	2,500	0	<0.1	50
E2	2.5	5	1	7,500	500	5	400
E3	5.0	10	2	10,000	1,000	10	800
E4	15	25	5	20,000	2,500	25	1000
* The values apply to both pre- and post-curfew, except in Zones 0 and 1 when the values shall be zero post-curfew. The values for signs do not apply to signs for traffic control purposes. # Up to 1 lux for public road lighting.							

Table 37: Environmental lighting zones

Zone	Lighting Environment	Surrounding	Examples
E0	Dark	Protected	UNESCO Starlight Reserves, IDA Dark Sky Parks
E1	Intrinsically dark	Natural	Relatively uninhabited rural areas, e.g. national parks

Zone	Lighting Environment	Surrounding	Examples
E2	Low district brightness	Rural	Sparsely inhabited rural areas
E3	Medium district brightness	Suburban	Well inhabited rural and urban settlements
E4	High district brightness	Urban	Town and city centres and other commercial areas

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
All	Nighttime lighting levels report or any other relevant study.
All	Photographic evidence confirming that external luminaires are designed to restrict upward light and light spill.
All	Confirmation that lighting is switched off during the curfew period.